

School of Electrical Engineering and Computer Science
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RESEARCH SUMMARY

My research focuses on data-efficient AI methods to accelerate scientific discovery and engineering design in domains where collecting data involves performing expensive experiments in physical lab or simulations. I specialize in surrogate modeling, and adaptive experimental design with high-impact applications in nanoporous materials discovery, additive manufacturing, and wearable devices for health monitoring. My work has been published at top-tier AI/ML conferences (ICML, AAAI, AISTATS)

EDUCATION

Washington State University, Pullman, WA 2022 – 2027 (Expected)
Doctor of Philosophy in Computer Science (GPA: 3.96/4.0)
Advisors: Prof. Jana Doppa and Prof. Nghia Hoang
Dissertation: *Use-inspired AI for Engineering Design and Scientific Discovery*

Washington State University, Pullman, WA 2022 – 2025
Master of Science in Computer Science (GPA: 3.97/4.0)
Advisors: Prof. Jana Doppa and Prof. Nghia Hoang
Thesis: *Nanoporous Materials Discovery via Search Bias-Guided Surrogate Modeling*

École Polytechnique de Tunisie, Tunis, Tunisia 2019 – 2022
Engineering Degree in Computer Science, Major: Signals and Systems
Graduated with Excellence

Preparatory School for Engineering Studies of Tunis (IPEIT), Tunisia 2017 – 2019
University First Cycle Studies, Dual Degree: Mathematics-Physics
Graduated with 76th National Rank

AWARDS AND HONORS

Founders' Award for Outstanding Masters Thesis Student, Association for Faculty Women 2026

Innovative Deployed AI Application Award, Association for the Advancement of Artificial Intelligence (AAAI) / IAAI 2026

AAAI Scholarship and Volunteer Award, Association for the Advancement of Artificial Intelligence (AAAI) 2026

Outstanding Graduate Teaching Assistant Award, Voiland College of Engineering, Washington State University 2025

Outstanding Graduate Teaching Assistant Award, EECS Department, Washington State University 2025

ICML Scholarship Award, International Conference on Machine Learning (ICML), Vienna, Austria 2024

Nakhara Scholarship, Voiland College of Engineering and Architecture, Washington State University 2022 – 2023

Tunisian Government Scholarship for Engineering Study, awarded for ranking in the *Top 5% nationally* in the Qualification Exam for Engineering Schools Entrance 2019 – 2022

PUBLICATIONS

CONFERENCE PAPERS

1. Minh Hoang, **Azza Fadhel**, Aryan Deshwal, Janardhan Rao Doppa, Trong Nghia Hoang. **Learning Surrogates for Offline Black-Box Optimization via Gradient Matching**. *Proceedings of International Conference on Machine Learning (ICML)*, 2024. (27% acceptance rate)
2. **Azza Fadhel**, Nathaniel W. Zuckschwerdt, Aryan Deshwal, Susmita Bose, Amit Bandyopadhyay, Jana Doppa. **Discovery of Feasible 3D Printing Configurations for Metal Alloys via AI-Driven Adaptive Experimental Design**. *Proceedings of AAAI Conference on Artificial Intelligence and Innovative Applications of AI Conference (AAAI/IAAI)*, 2026. (24% acceptance rate)
3. **Azza Fadhel**, Yassine Chemingui, Minh Hoang, Aryan Deshwal, Trong Nghia Hoang, Jana Doppa. **Nanoporous Materials Discovery via Search Bias-Guided Surrogate Modeling**. *Proceedings of AAAI Conference on Artificial Intelligence (AAAI), Artificial Intelligence for Social Impact Track*, 2026. (24.1% acceptance rate)
4. **Azza Fadhel**, The Hung Tran, Trong Nghia Hoang, Janardhan Rao Doppa. **Black-Box Optimization from Small Offline Datasets via Meta Learning with Synthetic Tasks**. *Proceedings of International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. (25% acceptance rate)
5. Nathaniel W. Zuckschwerdt, **Azza Fadhel**, Aryan Deshwal, Janardhan Rao Doppa, Susmita Bose, Amit Bandyopadhyay. **Artificial Intelligence Driven Approach to Accelerate Discovery of Directed Energy Deposition of GRCop-42**. *Journal of Virtual and Physical Prototyping*, 2026.

PAPERS UNDER REVIEW

1. The Hung Tran, **Azza Fadhel**, Quang Hung Pham, Manh Cuong Dao, Phi Le Nguyen, Jana Doppa, Trong Nghia Hoang. **Boosting Offline Reinforcement Learning in Data-Scarce Settings via Guided Diffusion Distillation**. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
2. Fatemeh Alizadehziri, **Azza Fadhel**, Ganapati Bhat, and Janardhan Rao Doppa. **Wear-DSE: Efficient Information-Theoretic Design Space Exploration Framework for Wearable Systems**. *Embedded Systems Week (ESWEEK)*, 2026.

RESEARCH EXPERIENCE

Graduate Research Assistant, Washington State University, EECS September 2022 – Present

- Developing machine learning-based surrogate models from experimental data for offline black-box optimization using gradient-matching.
- Created search bias-guided surrogate modeling frameworks combining value-matching and optimization-regularized objectives to accelerate discovery of high-performing materials.
- Designed meta-learning frameworks using Gaussian process-based synthetic task generation to learn transferable optimization bias and fine-tune surrogates under data scarcity.
- Built probabilistic surrogates for adaptive parameter search in metal additive manufacturing, enabling rapid, defect-free printing of NASA's GRCop-42 alloy.
- Developed multi-objective, multi-fidelity Bayesian optimization frameworks to jointly optimize energy efficiency, and sensing accuracy in healthcare wearables.

Research Intern, University of Louisville, USA March 2022 – May 2022

- Explored applications of quantum computing in machine learning and conducted comparative analysis of quantum computing results with traditional approaches.

Quantum Research Intern, QWorld (Online) June 2021 – August 2021

- Implemented hybrid quantum-classical models for image classification using Quanvolutional Neural Networks, Ensemble Classification, and Quantum Transfer Learning.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Washington State University, EECS January 2023 – Present

Courses:

- CptS 434/534: Neural Network Design and Applications (Spring 2023, 2025)
- CptS 451: Introduction to Database Systems (Spring 2025)
- CptS 350: Design and Analysis of Algorithms (Fall 2024)
- CptS 440/540: Artificial Intelligence (Spring 2024)
- CptS 570: Machine Learning (Fall 2023)

Responsibilities:

- Developed and delivered tutorials and hands-on labs on neural networks, SGD, fine-tuning, and CNNs/ResNets.
- Provided detailed feedback with clear rubrics, weekly office hours, and targeted help sessions to strengthen student understanding.

Section Leader, Code in Place, Stanford University (Online) April – May 2025

- Taught Python programming to a global cohort through live weekly sessions, emphasizing interactive learning and coding fundamentals.

Teaching Assistant (Volunteer), NVIDIA Deep Learning Institute April 2025

- Supported certified course labs on neural networks and PyTorch at Washington State University.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, SQL, Bash
- **Machine Learning & Deep Learning:** PyTorch, GPyTorch, BoTorch, scikit-learn, AutoGluon, Transformers (LLMs), Diffusion Models, Gaussian Processes, Bayesian Optimization, Surrogate Modeling, Meta-Learning
- **Data Science & Analytics:** NumPy, pandas, SciPy, Matplotlib, Seaborn; statistical modeling, experiment design, data visualization, feature engineering
- **Systems & Tools:** Linux, Git/GitHub, Conda, Slurm, VS Code, Jupyter, LaTeX/Overleaf
- **Languages:** Arabic (native), French (bilingual), English (professional)

PROFESSIONAL SERVICE

Workshop Organizer

- Co-Organizer, 5th AAAI Workshop on AI to Accelerate Science and Engineering (AI4Science), AAAI 2026. Collaborated with Aryan Deshwal, Jana Doppa, Vipin Kumar, and Carla Gomes.

Tutorial Organizer

- Co-instructor, Black-box Optimization from Offline Datasets: Foundations to Recent Advances, Half-day AAAI Tutorial (with Jana Doppa and Aryan Deshwal), AAAI 2026.

Conference Program Committee

- AAAI (2024, 2026), ICML (2025), AISI (2026), IAAI (2026), ICLR (2026), AISTATS (2026), ICML (2026), Neurips (2026)

Journal Reviewer

- IEEE Transactions on Sustainable Computing (2025)

Leadership & Mentoring

- Graduate Representative, GradSWE, Washington State University (Fall 2025 – Present): Organize the Women in Engineering Speaker Series to inspire and support women in STEM.
- Mentor, AI4ALL Remote Program (May – December 2025): Guiding undergraduate teams through hands-on AI projects; mentored 4 groups to date.
- QWorld (Summer 2021): Assisted participants with programming tasks and answering technical questions.

Volunteer Service

- Conference Volunteer: ICML (2024), WiML (2024), AAAI (2026)
- Judge, EECS Capstone Poster Competition, Washington State University (Fall 2025)

- Judge, CrimsonCode Hackathon, Washington State University (Spring 2026)

INVITED TALKS

Discovery of Feasible 3D Printing Configurations for Metal Alloys via AI-Driven Adaptive Experimental Design

AAAI Conference on Artificial Intelligence and Innovative Applications of AI Conference (AAAI/IAAI) 2026

Black-box Optimization from Offline Datasets: Foundations to Recent Advances

Half-day AAAI Tutorial (with Jana Doppa and Aryan Deshwal) 2026

PROFESSIONAL DEVELOPMENT

- **Advanced Technical Interview Prep**, CodePath, Spring 2026. Topics: data structures, algorithms, technical interview problem solving, and coding interview practice.
- **Participant – CRA-WP Mentoring Mentors Workshop Series 2025**, *Computing Research Association (Jun–Jul 2025)*:
5 Multi-session series on evidence-based mentoring practices; topics included inclusive mentorship, aligning expectations, and supporting mentee growth.
- **IEEE Women in Engineering International Leadership Summit (WIE ILS)**, *Tri-Cities, WA*:
Participated in workshops and panel discussions on leadership and career development, while networking with industry professionals.
- **Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)**, *tunis, Tunisia*:
Participated in the program "Promoting Female Talent" Improved soft skills, expanded the professional network, and explored the real-world business environment.